RESEARCH METHODOLOGY REPORT

Methodological Reconstruction and Critical Analysis of HAPPO-Based Joint Operation of Five Cascade Reservoirs

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A reconstruction of the implemented physical model, multi-agent decision formulation, HAPPO optimization, experimental workflow and evidence-supported limitations.

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Scope: supplied inputs, static implementation, configurations and existing experimental records.

No training, simulation, evaluation, project-module execution or new scientific experiment was performed for this report. Equations reconstructed from executable logic are marked R; theoretical comparisons and proposed changes are identified separately.

Prepared for researchers in reservoir optimization, hydropower operation and multi-agent reinforcement learning.

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Chapters are organized by scientific operations. Evidence identifiers in square brackets resolve to the traceability appendix. Page references are generated from the final document layout.

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Reading conventions

V: verified existing implementation or archived record. **R:** reconstructed mathematical interpretation. **T:** standard theoretical background. **I:** confirmed implementation issue. **C:** evidence-supported concern. **H:** hypothesis requiring verification. **F:** proposed future improvement. Verification of implementation does not constitute physical calibration or experimental reproduction.

1. Executive Summary

The repository implements a simplified daily physical model of five cascade reservoirs and learns five independent categorical operating policies through HAPPO-style sequential updates. The actors choose positions within state-dependent release intervals. A centralized physical mechanism constructs releases compatible with current-day storage and outlet limits; generation and service deficits then determine reservoir-specific rewards.

The default task has 1,825 daily decisions, retained leap days, prescribed historical incremental inflows and supplied initial storages. The physical chain is Wudongde (WDD), Baihetan (BHT), Xiluodu (XLD), Xiangjiaba (XJB), and Three Gorges (THR). All upstream release reaches the next reservoir within the same step. The model does not represent river travel time, evaporation, seepage, withdrawals or unit-level operating decisions.

Five actors receive local and immediate-neighbor features. A centralized six-output critic estimates five mixed returns and one retained legacy return. Each actor learns from 50% local penalized normalized generation and 50% global normalized generation. The displayed legacy return and separately reported system reward have different definitions. Neither should be treated as interchangeable with total energy or the sum of actor returns.

The default training budget is 300 complete episodes per seed, collected six at a time, giving 50 policy-update rounds and 547,500 environment transitions. Policies remain frozen during collection. GAE and per-channel advantage normalization precede randomized sequential actor updates; critic optimization follows. Periodic evaluation uses three fixed training-only windows. Final evaluation uses the last policy on one concatenated sequence of held-out years.

Five archived experiments were inspected. Their settings and input/source versions differ, so the collection does not constitute a controlled ablation series. Three seeds provide descriptive variability. Existing outputs support internal feasibility, reward-accounting and saved-trajectory arithmetic checks; no new model run or physical calibration was performed.

Principal methodological implications

The most consequential concerns are the distinction between learning and reported objectives; strong episode-start discounting over a long scheduling horizon; absent terminal water valuation; limited anticipatory information; a state-dependent discrete action map; and incomplete physical and comparative validation. Current late-stage rewards are not generally penalty-dominated. Rare forced actions and low entropy alone do not establish the cause of unsatisfactory performance.

The report reconstructs what is implemented and specifies what further evidence would be needed to establish engineering fidelity, causal improvements or superiority over alternative scheduling methods.

Evidence: [P1-P6], [D1-D4], [M1-M5], [A1-A6], [X1-X4]. Source-based facts and derived expressions are separated throughout.

2. Research Background, Scope, and Problem Formulation

2.1 Scientific decision and scheduling horizon

The operational problem is to allocate release and storage across hydraulically connected reservoirs while producing hydropower and responding to prescribed output and ecological service requirements. Daily releases affect present generation, downstream water availability, future storage and head. Learning is used to select operating commands under these coupled consequences.

The audited default episode contains T = 1,825 decisions with a fixed 86,400-second step. The episode horizon is configurable; the daily physical step is enforced. A training episode begins on 1 January and uses consecutive source days. Leap days remain; therefore a nominal five-year episode can end before 31 December of its fifth source year. No within-day dispatch or price variation is modeled.

2.2 Mathematical reconstruction of the learning criterion

$$J_i(\theta) = E[\sum_{t=0}^{T-1} \gamma^t r_{i,t}], \; i = 1,\ldots,5. \tag{1}$$

Here θ denotes the collection of five policy parameter sets; r is the actor-specific mixed reward defined in Chapter 8; and γ is the configured discount. The expectation summarizes command sampling and exposure to scheduled training windows. The actual finite sampling schedule need not be independent and identically distributed. HAPPO performs approximate policy improvement using sampled surrogates; it does not solve this criterion exactly.

The physical decisions are total releases Q (m^3/s), with resulting storages V (m^3), powers P (MW) and energies E (MWh). The learned decision variables are categorical commands k. Turbine discharge, spill and power are derived rather than independently optimized.

2.3 Criteria, trade-offs and scope

Local generation is normalized by each reservoir capacity, while global generation is normalized by aggregate installed capacity. Unit-weight relative output and flow shortfalls compete with generation. These are explicit dimensionless trade-offs, not monetary costs or calibrated social preferences. The five actors target distinct mixed returns; one common cascade return is not supplied to all actors.

Hard conditions include water balance, operating storage envelopes, total-release capacity and modeled turbine/power limits. Service minima remain soft. No terminal restoration or residual-water value is included. No verified electricity-price benefit, flood-damage objective, navigation term or explicit spill penalty is present in the investigated training path.

The author-intended external optimization standard beyond these implemented criteria is **Not identifiable from the available repository.** Consequently, assessment must distinguish successful optimization of a learning signal from the engineering desirability of the resulting schedule.

Evidence: Equation (1): R, from rewards and rollout losses [M3, A2-A4]. Horizon and effective defaults: V [D2, A6].

3. Physical Configuration of the Five-Reservoir System

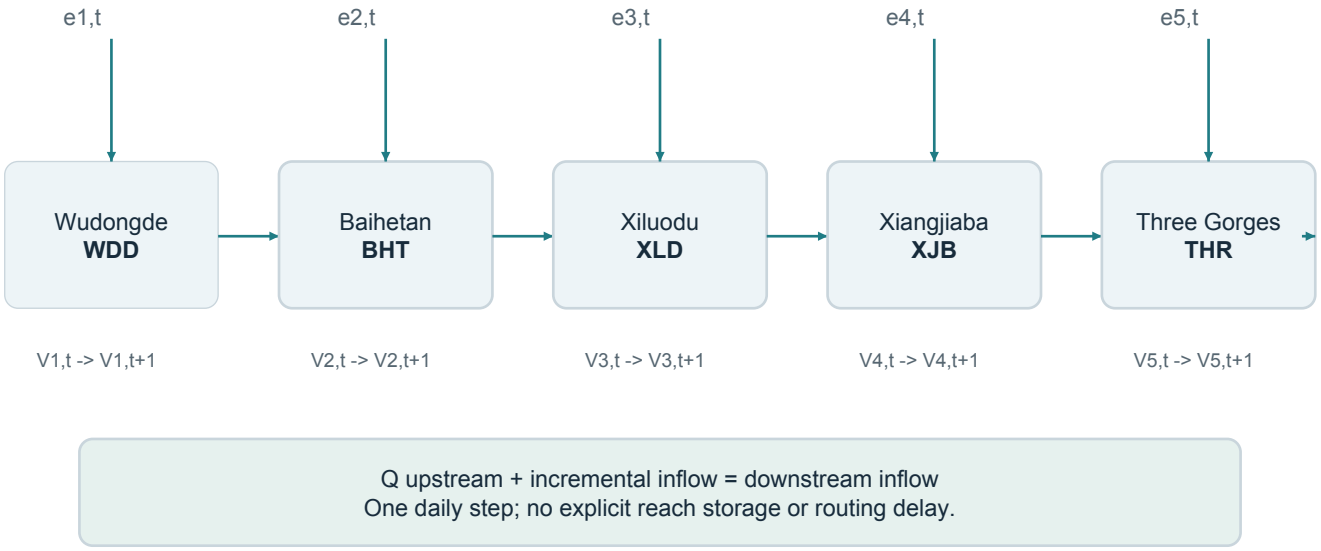


Figure 1. Implemented hydraulic topology. Incremental external inflow e enters each reservoir; total upstream release is transferred to the next reservoir in the same daily step. The drawing is a model schematic, without geographic scale or verified travel times.

Reservoir	Elevation m	Storage 10^9 m³	Installed MW	Nominal turbine m³/s	Total outlet m³/s
WDD	945-975	2.843-5.863	10,200	8,080	35,000
BHT	765-825	8.570-19.006	16,000	8,600	42,348
XLD	540-600	5.110-11.570	12,600	7,500	52,300
XJB	370-380	4.074-4.977	6,400	7,056	48,660
THR	145-175	17.150-39.300	22,500	31,885	102,500

Table 1. Configured global operating ranges and supplied engineering capacities. Storage limits are obtained consistently from the supplied curves. The loader supports explicit installed-capacity overrides; this table reports the audited defaults.

The total installed capacity is 67,700 MW. Different storage ranges, hydraulic heads and outlet capacities create physical heterogeneity. Independent policy parameters accommodate reservoir-specific behavior, although all actors have the same network structure and number of categorical commands.

There is no separately represented tributary network or intervening reservoir. Incremental inflows summarize external contributions between modeled nodes. The model contains no explicit reach storage, propagation delay, attenuation or loss between reservoirs. These are verified simplifications, not findings about actual river hydraulics.

Evidence: Topology and flow transfer [P1]; values [P2]; operating domains and curve conversion [D1]. THR maps to TGR in the service-source attribution [D4].

4. Reconstructed Mathematical Formulation of Cascade Operation

4.1 Connectivity and water balance

$$I_{1,t} = e_{1,t};\ I_{i,t} = e_{i,t} + Q_{i-1,t}\ (i > 1). \quad (2)$$

$$V_{i,t+1} = V_{i,t} + \Delta t\ (I_{i,t} - Q_{i,t}),\ \Delta t = 86{,}400\ \text{s}. \quad (3)$$

V is beginning/resulting storage in m^3; I, e and Q are daily mean rates in m^3/s. Multiplication by seconds converts net rate to daily volume. For WDD, external inflow is its complete modeled inflow. For downstream nodes, e is an interval contribution and must not already include the same upstream release.

$$\sum_{i=1}^{5}(V_{i,t+1} - V_{i,t}) = \Delta t\ (\sum_{i=1}^{5} e_{i,t} - Q_{5,t}). \quad (4)$$

Equation (4) is a derived cascade accounting identity: internal releases cancel. Only external contributions and final outlet discharge cross the modeled system boundary. This identity underlies an explicit numerical consistency check. It does not validate the supplied forcing against measured basin water balance.

4.2 Reservoir geometry

$$Z_i(V) = z_j + (V-v_j)(z_{j+1}-z_j)/(v_{j+1}-v_j), \quad (5)$$

for adjacent supplied storage/elevation anchors (v_j, z_j) bracketing V. Elevation Z and z are in m; v and V are in m^3. The inverse conversion also uses piecewise linear interpolation. No analytical bathymetric polynomial is fitted during the operating calculation.

Reservoir	All anchors	Anchors within operating domain	Full curve versus operating domain
WDD	84	61	Curve extends above 975 m
BHT	136	121	Curve extends above 825 m
XLD	121	121	Same supplied domain
XJB	21	21	Same supplied domain
THR	72	61	Curve extends above 175 m

Table 2. Existing curve-anchor counts. Counts do not establish survey accuracy. The full interpolation domain is not permission to operate above configured ceilings.

Evidence: Equations (2)-(5): R, directly supported by [P1, D1]. Curve and balance assumptions are V; original measurement provenance remains unresolved.

4. Mathematical Formulation: Hydraulic Head and Generation

4.3 Assumed storage-sensitive net head

$$Z_i^{ref} = (Z_i^{min}+Z_i^{max})/2;\ Z_{i,t} = [Z_i(V_{i,t})+Z_i(V_{i,t+1})]/2;$$
$$H_{i,t} = H_i^{ref} + s_i(Z_{i,t}-Z_i^{ref}),\ s_i=1. \quad (6)$$

H is net head in m. The beginning/end *elevations* are averaged; this is not evaluation at mean storage or a resolved daily hydraulic integration. A constant combined tailwater/loss offset is assumed. The reference midpoint is an assumption, not a measured reference condition. Separate tailwater and loss components remain unknown.

Reservoir	Z ref / m	H ref / m	Combined offset / m	Scenario head range / m
WDD	960	143	817	128-158
BHT	795	202	593	172-232
XLD	570	185	385	155-215
XJB	375	100	275	95-105
THR	160	80.6	79.4	65.6-95.6

Table 3. Assumed head-sensitivity scenario. Every supplied row is marked assumed_sensitivity_only. Fixed-head mode instead uses H = H ref and is an available alternative, not the current default.

4.4 Turbine allocation, power and daily energy

$$c_{i,t} = \rho g \eta H_{i,t}/10^6;\ q_{i,t}^{tur} = \min(Q_{i,t}, q_i^{nom}, P_i^{cap}/c_{i,t}). \quad (7)$$

$$P_{i,t} = c_{i,t} q_{i,t}^{tur};\ q_{i,t}^{spill} = Q_{i,t}-q_{i,t}^{tur};\ E_{i,t} = 24P_{i,t}. \quad (8)$$

ρ = 1,000 kg/m^3, g = 9.81 m/s^2 and η = 0.90. The factor 10^6 converts W to MW; E is MWh. The effective turbine cap simultaneously respects nominal turbine discharge and installed power at the modeled daily mean head. Excess total release becomes spill automatically. There is no separate turbine/spill decision or unit-efficiency curve.

Installed capacity is enforced for daily mean modeled output, not an explicitly resolved instantaneous trajectory. Additional release need not increase power under the head-sensitive model because drawdown changes head. The reference cascade daily energy is 24 × 67,700 = 1,624,800 MWh.

Evidence: Equations (6)-(8): R, corroborated by the documented sensitivity assumptions [P3-P4]. No measured/calibrated hydraulic relationship is claimed.

4. Mathematical Formulation: Joint Release Feasibility

4.5 Backward compatibility, then forward physical execution

Let $a_{i,t}$ and $b_{i,t}$ be lower/upper storage limits on the state reached after today's action. Suppress t below. Before applying commands, the mechanism computes the total releases that admit a feasible downstream completion. Initialize $L_5^d = 0$ and $U_5^d = Q_5^{cap}$. For i = 4,...,1:

$$L_i^d = \max[0, L_{i+1}^d - e_{i+1} + (a_{i+1} - V_{i+1})/\Delta t];$$
$$U_i^d = \min[Q_i^{cap}, U_{i+1}^d - e_{i+1} + (b_{i+1} - V_{i+1})/\Delta t]. \quad (9)$$

These are downstream-compatible intervals, before imposing reservoir i's own storage balance. Requiring the next reservoir's balance-derived release interval to intersect $[L_{i+1}^d, U_{i+1}^d]$ yields the two inequalities in (9). Every term has flow units.

The mechanism then executes WDD through THR. Once actual upstream release is known, total inflow is determined and the local interval becomes

$$L_i = \max[L_i^d, I_i + (V_i - b_i)/\Delta t];$$
$$U_i = \min[U_i^d, I_i + (V_i - a_i)/\Delta t]. \quad (10)$$

$$u_i = k_i/(K-1); \; Q_i = L_i + u_i^2 (U_i - L_i), \; k_i = 0,...,K-1. \quad (11)$$

The quadratic map spans both endpoints and concentrates command resolution near the lower endpoint. With K = 100, each actor samples one of 100 normalized positions; different days and upstream commands produce different physical grids.

4.6 Degenerate and infeasible cases

An empty interval beyond 10^{-8} m^3/s triggers failure. A width at most that tolerance causes forced midpoint release, bounded to the physical outlet range. The recorded normalized value 0.5 is not squared in this branch; the effective categorical index is a placeholder zero.

In real arithmetic, this backward/forward construction characterizes single-day continuous feasibility for the implemented linear no-lag chain with storage and outlet bounds. It is not a proof of multiday viability, service satisfaction, grid adequacy or real routing accuracy.

Evidence: Equations (9)-(11): R from [P5]. Single-day derivation is an interpretation of interval-intersection logic. No future forcing or deadline trajectory is used.

4. Mathematical Formulation: Constraints and Boundaries

Condition	Implemented treatment	Scope / interpretation
Water balance	Hard transition and residual checks	Storage is computed, not projected onto bounds.
Global storage/elevation	Hard operating domain	Derived consistently from supplied curve and configured elevations.
Dated end-state envelope	Hard, via feasible release intervals	Date t row constrains storage after action t.
Total release	Hard 0 to outlet capacity	Separate from turbine capacity.
Turbine / installed output	Automatic flow allocation and cap	Daily mean head and power; no unit commitment.
Inter-reservoir coupling	Hard same-day flow and joint feasibility	All total upstream release enters next reservoir.
Guaranteed power / ecology	Soft relative-shortfall penalties	Neither is a minimum hard constraint.
Ramp rates / level changes	No independent ramp constraint identified	Changes restricted indirectly by balance and envelopes.
Flood control	Only supplied daily envelopes identifiable	Regulatory provenance and damage objective unresolved.
Initial / terminal storage	Initial table; final daily envelope	No end restoration, salvage value or future viability constraint.

Table 4. Constraint classification. Absence of a separately modeled constraint is a scope fact, not automatically an implementation defect.

4.7 Timing, numerical tolerances and conservation

Current storage is checked against the preceding executed row; reset checks global bounds and sets previous release to zero. Year joins carry storage and previous release. The row for action date t is used directly, without an extra one-day shift. The last action's row constrains terminal storage.

Storage tolerance is generally max(1 m^3, $10^{-10}|V|$); transition-start global checking separately permits ±1 m^3. Flow tolerance is 10^{-8} m^3/s and the installed-power assertion permits 10^{-7} MW. Physical storage is not clipped. Elevation-lookup input may be clipped within accepted numerical tolerance.

The flow-volume conversion, 24-hour energy conversion and daily input timing are compatible across all five reservoirs. The saved-trajectory arithmetic audit in Appendix D supports numerical consistency. Conservation does not verify head calibration, routing delay, input authenticity or future operational sustainability.

Evidence: V [P1, P5-P6, D2-D3]; tolerance details and scoped numerical audit [X4].

5. Data Sources, Preprocessing, and Experimental Assumptions

Input	Inspected extent	Scientific role
Daily incremental inflows	21,915 dates; 1965-2024	Daily external forcing for five nodes.
Initial storage	300 reservoir-year records	One initial vector per episode starting year.
Stage-storage anchors	434 total anchors	Linear geometry and inverse conversions.
Engineering parameters	Five reservoir rows	Capacity, reference head and operating assumptions.
Head sensitivity	Five assumed-scenario rows	Storage-to-head feedback, not calibration.
Service requirements	Five reservoir rows	Base and effective output/flow thresholds.
Dated operating bounds	109,575 reservoir-date rows	End-state envelope for every action date.

Table 5. Supplied input inventory. Counts reflect inspected files, not independent provenance validation.

5.1 Validation and normalization

The loading path checks finite values, nonnegative inflows/thresholds, date coverage, reservoir identity, monotone geometry, parameter compatibility and storage/elevation bound consistency. It does not impute missing hydrology, resample time series, smooth flows, generate forecasts or fit a hydrological model. Original preparation outside this path is not inferred.

For each reservoir, standardized external inflow is $(e - \mu_{train})/\sigma_{train}$, with the standard-deviation denominator floored at 1. The 55 designated training years supply 20,090 dates for this calculation. The 39 eligible training windows cover 20,084 unique dates; six unused boundary dates still enter training-only normalization. This is not held-out leakage.

5.2 Temporal split and experimental assumptions

Held-out years are [1998, 1999, 1970, 1997, 2006]; all other source years are training years. The split isolates designated years but is not a chronological forecast-to-future split. Training windows start on 1 January, contain 1,825 consecutive days and contain no held-out date.

Coverage-balanced sampling approximately equalizes source-year-day exposure, creates integer quotas and uses six distinct windows per batch. It does not balance hydrological labels or guarantee exactly equal exposure. A uniform-with-replacement alternative exists.

Daily forcing is assumed known before action selection. Initial conditions vary with the selected start year. Original hydrology, initial-storage construction, curve surveys and envelope-generation procedures are **Not identifiable from the available repository.** The service paper's identity is corroborated, but its exact Table A.1 values/page attribution remain partially verified.

Evidence: V [D1-D4]. Coverage balancing is implemented, not evidence of hydrologically independent samples or representativeness.

6. Multi-Agent Reinforcement Learning Formulation

6.1 Five agents and coupled consequences

One independently parameterized actor is assigned to each reservoir. Physical heterogeneity comes from reservoir-specific geometry, storage, head and capacities; policy heterogeneity comes from separate weights. All actors nevertheless use the same MLP architecture and K-bin command representation. A command changes own storage and generation and can change downstream inflow, feasible releases and future operating conditions.

The task is a finite-horizon partially observed multi-agent control problem with agent-specific mixed returns. Calling it a common-reward Dec-POMDP would conceal the different actor rewards. A conceptual latent state includes the selected forcing sequence, its current position, storage and previous release. The practical centralized feature vector is not proven to be a sufficient Markov state.

6.2 Centralized training and operational execution

During training, one critic receives system-wide features and predicts six return channels. Each actor receives local and immediate-neighbor information. Joint commands are sampled before today's upstream releases have been resolved. The environment then uses all reservoir states, inflows and end-state limits to construct actual releases.

This supports CTDE at the policy/value architecture level. Actual dispatch still depends on neighbor data and a system-wide feasibility mechanism. The implementation does not establish fully autonomous local-only control. The critic does not choose actions, compute optimal releases, provide a counterfactual Q-baseline or calculate each agent's marginal contribution.

Element	Implemented interpretation
Agent	One reservoir-specific command policy.
Joint command	Five categorical indices; conditional policy sampling factorizes.
Transition	Deterministic given the episode forcing and commands.
Actor information	15 current local/neighbor/rule/time features.
Training information	38-feature centralized input; six value outputs.
Reward	Own local/global mixed reward; common global generation component.
Horizon	True finite episode; terminal bootstrap masked out.

Table 6. Decision-problem interpretation. Factorized command sampling does not imply factorized physical releases; the shared feasibility map couples execution.

No recurrent state, attention module, learned communication, policy parameter sharing or differentiation through the physical transition is present in the investigated path. These are descriptive scope statements rather than prerequisites that HAPPO must satisfy.

Evidence: V [M1-M2, A1-A2]. Partially observed control interpretation is R; complete Markov sufficiency and common-objective guarantees remain unestablished.

7. State, Observation, and Action Space Design

7.1 Exact default actor observation

Index	Feature	Physical/time interpretation and scaling
0	Own storage	(V - global V min) / global storage span.
1	Own external inflow	Today's supplied interval inflow, training z-score.
2	Own previous release	Yesterday's actual total release / own outlet capacity.
3-5	Immediate upstream	Its storage fraction, today's external z-score, yesterday's release fraction.
6	Downstream storage	Immediate downstream reservoir's storage fraction.
7	Downstream headroom	(Today's end-state upper storage - current downstream storage) / its global storage span; signed.
8-9	Current own bounds	Lower/upper allowed beginning-state storage, normalized by global storage range.
10-11	Own end-state limits	Date-t row's resulting-state level lower/upper limits, normalized by global level range.
12-13	Season	Sine/cosine of 2π(day-of-year - 1)/days-in-source-year.
14	Remaining time	(T - t)/T; one at episode start.

Table 7. All 15 features, in exact order. Boundary placeholders are upstream (0,0,0) for WDD and downstream (0,1) for THR. Observations are float32, are not clipped to [0,1], and have unbounded declared Box limits.

7.2 Centralized features and information limits

The 38 critic inputs concatenate five storage fractions, five external-inflow z-scores, five previous-release fractions, three time features, five current lower and five current upper storage bounds, and five end-state lower and five end-state upper level limits. One vector is used per lane/day; it is not a concatenation of five copies into 190 inputs.

With supplied envelopes disabled, actors receive 11 features and the critic 18; the four/20 boundary features are omitted. This optional configuration is not the default.

Today's external daily mean is supplied before action selection. Today's actual upstream release is unavailable to actors; yesterday's release is used. Neither future inflow trajectories, explicit source-year identity, full system state, service thresholds nor head are direct actor inputs. Separate actor weights implicitly identify the reservoir. Current inputs have globally fixed lower envelopes, so some boundary features are constant.

Storage encodes past net water allocation, but feed-forward policies have no learned recurrent memory. Seasonal features may support statistical anticipation; absence of forecasts does not prove inability to anticipate, and absence of year identity alone does not prove non-Markov observations.

Evidence: V [M2]. Operational availability of exact current-day means and sufficiency for long-term coordination remain C/H, not verified engineering assumptions.

7. State, Observation, and Action Design: Commands

7.3 From probabilities to categorical commands

Each actor maps its 15-vector through four hidden layers [64,128,256,256] to K logits. Hidden blocks include ReLU and LayerNorm; disabled input feature normalization does not remove hidden LayerNorm. Orthogonal initialization is used; the actor output gain is 0.01. A categorical distribution provides sampled commands and their log probabilities.

Training samples k from that distribution; deterministic evaluation selects the most probable index, not an expected index or mean physical discharge. The sampled joint command distribution factorizes conditionally on observations, but the realized release vector is coupled by the physical mechanism.

Quantity	Domain	Interpretation
Logits / categorical probabilities	K real logits / probabilities summing to 1	Actor output before physical conversion.
Sampled command k	Integer 0,...,K-1	Default K=100; older archives K=200.
Normalized command u	k/(K-1), in [0,1]	A dimensionless interval position.
Mapping fraction	u^2	Nonuniform release-grid coordinate.
Actual total release Q	$L + u^2(U-L)$	Physical m^3/s; depends on state and upstream execution.
Forced branch	Width at most 10^-8 m^3/s	Midpoint flow; actor control mask zero.

Table 8. Command variables and physical decisions. All bins are available in the ordinary path. Integer/range validation rejects invalid commands; there is no Gaussian policy, tanh conversion or ordinary post-hoc command clipping.

7.4 Consequences for learning and feasibility

For a nondegenerate interval, increasing k increases release. Its physical increment depends on both k and interval width. The lower region has finer resolution; large feasible intervals can still give coarse physical increments at high k. A command's meaning changes as storage, external forcing, bounds or upstream commands change.

The feasibility mechanism is part of the operational model rather than a differentiable policy layer. Policy gradients use command likelihoods and observed returns; no derivative of the balance, head or interval endpoints is propagated through the physical model.

A forced row retains its physical reward and transition for value learning, while actor loss and entropy contributions are masked. The sampled index, effective placeholder index, normalized bookkeeping value and actual flow must be distinguished when reading diagnostics. The observed rarity of forced decisions does not establish universal grid adequacy.

Evidence: V [M1-M2, P5, A2-A3]. Resolution limitations are C; their actual performance impact has not been isolated.

8. Reward Design and Constraint Handling

8.1 Service deficits and physical generation terms

$$d_{i,t}^{P} = \max(G_i - P_{i,t}, 0)/G_i;\ d_{i,t}^{F} = \max(F_i - Q_{i,t}, 0)/F_i. \quad (12)$$

G is an effective guaranteed-output threshold in MW and F an effective ecological total-release threshold in m^3/s. These divisions apply only to positive thresholds; an inactive threshold has normalized deficit zero. Both deficits are dimensionless. Zero requirements are inactive; all current requirements are positive. Raw shortfalls no larger than 10^{-10} max(threshold,1) are ignored in both penalties and counts.

Reservoir	Base G / MW	Effective G / MW	Base F / m³/s	Effective F / m³/s
WDD	2,290	916	900	360
BHT	4,567.5	1,827	1,160	464
XLD	5,345.7	2,138.28	1,160	464
XJB	2,750	1,100	1,200	480
THR	4,990	1,996	6,000	2,400

Table 9. Reference CSV thresholds and current effective thresholds. Both scale factors are 0.4 and apply once to base values. A YAML comment still describes 0.5/0.7; executable values take precedence.

$$\ell_{i,t} = P_{i,t}/P_i^{cap} - \chi(w_P d_{i,t}^{P} + w_F d_{i,t}^{F});$$
$$g_t = \sum_i P_{i,t} / \sum_i P_i^{cap};\ r_{i,t} = 0.5\ell_{i,t} + 0.5 g_t. \quad (13)$$

χ is the penalty-enabled switch; current $\chi=1$ and $w_P=w_F=1$. Local normalized generation treats reservoir utilization individually. Global generation weights actual power by aggregate installed capacity and includes the actor's own output. It is also equal to total daily energy divided by the reference daily energy.

Power requirements compare daily mean MW, while ecology compares total outlet flow including spill. Neither enters the hard release interval. Constant thresholds apply through all dates; no seasonal gating is used. Negative rewards are retained without clipping.

No navigation, monetary, direct spill, action-correction, final-storage or residual-water term is included. Entropy is a separate optimization regularizer, described in Chapter 9.

Evidence: Equations (12)-(13): R [M3]. Effective/base values V [D4]. The exact external Table A.1 extraction remains partially verified.

8. Reward Design: Accounting, Trade-offs and Constraints

8.2 Three distinct return definitions

$$r_t^{system} = 0.5\sum_i \ell_{i,t} + 0.5 g_t;$$
$$r_t^{legacy} = g_t - \chi[w_P \text{mean}_{\text{active } P}(d^P) + w_F \text{mean}_{\text{active } F}(d^F)]. \quad (14)$$

$$R_c = \sum_{t=0}^{T-1} \gamma^t r_{c,t}; \; \sum_i r_{i,t} = r_t^{system} + 2g_t. \quad (15)$$

c identifies a reward channel. The legacy reward averages each penalty over its fixed positive-requirement set, not just violating reservoirs. The local reward does not divide its own deficits by five. The system scalar counts global generation once; summing actor mixed rewards counts it five times with coefficient 0.5. The critic retains legacy return as a sixth trained output, while actors use their own mixed channels.

Term or condition	Where it acts	Scientific consequence
Own utilization + global generation	Actor environment reward	Related but nonidentical actor incentives.
Relative power/flow deficit	Soft reward penalty	Compliance is encouraged, not guaranteed.
Storage/outlet feasibility	Central physical execution	Daily hard compliance primarily enforced by model.
Entropy coefficient	Actor loss only	Exploration regularization, not an extra service benefit.
Episode-start discount	Returns and actor/entropy row weights	Later calendar decisions carry smaller objective weights.
Terminal water	No explicit reward or equality	Future value ends at the last modeled day.

Table 10. Reward and feasibility roles. Reward weights are prespecified dimensionless trade-offs; no cost calibration is identified.

8.3 Alignment and magnitude

Under unit weights, loose algebraic daily bounds are local [-2,1], mixed [-1,1], legacy [-2,1] and system [-5,3]; these are not claims that every bound is jointly attainable. Physical energy and service frequencies are undiscounted metrics. Maximizing a discounted mixed return need not maximize total energy, minimize violation frequency or maximize the legacy curve.

At γ=.99, the discount half-life is 68.97 days, and the first 365 days contain 97.448% of the 1,825-day pure time-weight mass. This is not a measured gradient fraction, a strict planning horizon or proof of erroneous double discounting.

Evidence: Equations (14)-(15): R [M3, A3]. Archived decompositions [X2] show that reference and relaxed-threshold runs have different penalty balances.

9. HAPPO Architecture and Optimization Mechanism

9.1 Networks and effective defaults

Five independent actor MLPs map 15 observations to K categorical logits. A separate centralized MLP maps 38 inputs to six linear value outputs, with a shared trunk. Its first five heads predict own mixed returns; the sixth predicts legacy return and participates in the same critic loss. There are no recurrent, attention or communication modules.

Setting	Audited default / mechanism
Hidden widths	[64,128,256,256], actors and critic separately.
Hidden operations	Linear -> ReLU -> LayerNorm; input feature normalization disabled.
Initialization	Orthogonal; hidden ReLU gain; actor output gain .01; zero biases.
Policy / value outputs	Categorical K=100 / six linear normalized values.
Parameter sharing / agent order	No sharing / random permutation per update.
Rollout / budget	6 complete 1,825-day episodes / 300 episodes per seed.
Gamma / GAE lambda	.99 / .95.
Actor / critic epochs	5 / 5; one full-batch minibatch each epoch.
PPO clip / entropy coefficient	.2 / .001.
Actor / critic learning rate	.0003 / .0003, fixed by default.
Adam epsilon / weight decay	.00001 / 0.
Gradient norm bound	.5 for both networks.
Value loss	ValueNorm; half squared error; coefficient 1.
Clipped value / Huber loss	Both disabled; clipped value plus ValueNorm rejected.
KL controls	Local target null; rollback false; joint target null.
Execution / seeds	CPU threads 1; deterministic settings requested; seed controls applied.

Table 11. Current file defaults. Archived effective overrides take precedence over defaults. The YAML lists seeds 1-5; all five investigated archives actually record seeds 1,2,3.

9.2 Role of the critic

The critic supplies a state-dependent expected-return baseline for advantage estimation. It is not an action optimizer. Six-output training permits channel-specific values with shared features; the retained legacy target can influence the shared representation even though no actor directly uses its advantage. Harm or benefit of that influence has not been isolated.

Evidence: V [A1, A6]. Default settings are distinguished from optional controls and archived configurations in Chapter 12.

9. HAPPO: Trajectory Targets and Sequential Updates

9.3 Frozen on-policy collection and GAE

Each batch contains B = 1,825 × 6 = 10,950 environment rows, collected under fixed actor/critic parameters. Actor buffers keep observations, effective commands, rollout log probabilities and controllability masks. The critic buffer keeps one centralized vector per lane/day plus five mixed rewards and one legacy reward. There is no off-policy replay buffer.

$$\delta_{t,c} = r_{t,c}+\gamma m_{t+1} v_{t+1,c}^{old}-v_{t,c}^{old};$$
$$\hat{A}_{t,c} = \delta_{t,c}+\gamma\lambda m_{t+1}\hat{A}_{t+1,c};\ Y_{t,c} = \hat{A}_{t,c}+v_{t,c}^{old}. \quad (16)$$

v denotes stored predictions denormalized using collection-time ValueNorm statistics. c indexes six reward channels, not reservoirs alone. m is one except zero after the final transition; bad-transition masks are one. The final reward therefore has no terminal bootstrap. Year boundaries do not reset the GAE recursion. Y is the fixed GAE λ-return target for this update.

$$\tilde{A}_{b,c} = (\hat{A}_{b,c}-\mu_c)/(\sigma_c+10^{-5}). \quad (17)$$

Default moments in (17) use all common rollout rows separately per channel, including forced rows. Each actor receives its own normalized channel. Disabling the shared-normalization flag changes normalization scope to actor-controllable rows; it does not eliminate normalization.

9.4 Sequential policy correction

$$F_b^{(k)} = \prod_{j\,\text{before}\,i} \bar{\rho}_{j,b};$$
$$\bar{\rho}_{j,b} = \pi_j^{after}(k_{j,b}|o_{j,b})/\pi_j^{before}(k_{j,b}|o_{j,b}). \quad (18)$$

The five actors are updated in a random permutation. F begins at one and multiplies the completed actor's after/before likelihood ratio on the same stored commands. Because actors have separate parameters, an unupdated actor's before policy equals its collection policy. Forced-row ratios are replaced by one. F is detached, unclipped and checked for finiteness.

Advantages and F remain fixed during one actor's epochs. The next actor uses the updated product; no new state trajectory, reward or advantage is collected between actors. This addresses policy changes statistically within a fixed batch rather than recomputing a physical counterfactual.

Evidence: Equations (16)-(18): R [A2-A4]. GAE is an approximation using learned values, not measured true advantages.

9. HAPPO: Policy and Value Optimization

9.5 Actual clipped actor objective

$$\rho_{i,b} = \exp[\log \pi_{\theta i}(k_{i,b}|o_{i,b}) - \log \pi_i^{rollout}(k_{i,b}|o_{i,b})];$$
$$L_i = -\langle d_b C_{i,b} F_b \min[\rho_{i,b}\tilde{A}_{i,b}, \mathrm{clip}(\rho_{i,b}, 1-\varepsilon, 1+\varepsilon)\tilde{A}_{i,b}]\rangle_B - \beta_H \langle d_b C_{i,b} H(\pi_i)\rangle_B. \quad (19)$$

$d_b=\gamma^{\tau b}$ is the episode-start time weight; C is the controllability mask; $\varepsilon=.2$; $\beta_H=.001$; and angle brackets are an arithmetic mean over all rows. F is outside clipping; entropy is not multiplied by F. The denominator is neither active count nor discounted weight mass. Equation (19)'s probability ratio describes controllable rows; forced rows store a placeholder index and zero old log-probability and are masked from this objective. An all-zero objective batch skips Adam.

Rollout probabilities remain the PPO denominator across epochs. Ratios may lie outside [0.8,1.2]; clipping modifies the surrogate and is not a hard probability-ratio constraint. GAE discounting forms reward-to-go, while d weights episode-start occupancy. Their coexistence is not itself an implementation error.

9.6 Critic and normalization chronology

$$L_V = [1/(12B)] \sum_{b=1}^{B}\sum_{c=1}^{6}[N_{vnew}(Y_{b,c}) - v_{\psi,c}(x_b)]^2. \quad (20)$$

After all actors update, the critic forwards the centralized features, updates ValueNorm with fixed Y, normalizes targets with the new moments, backpropagates half squared error over all six channels, clips gradients and applies Adam. Five full-batch epochs update moments five times on the same targets. No additional actor time weights or controllability masks enter this value loss.

ValueNorm uses debiased exponential first/second moments ($\beta=.99999$), variance floor .01 and no PopArt output-layer compensation. Before/after denormalized fit changes combine network and normalization changes. Forward-before-moment-update is not by itself proof of an invalid loss: the network output does not directly read the moments.

9.7 Theory and optional modifications

Sequential factors and PPO clipping are implemented HAPPO mechanisms [1,2]. The original common-return theory cannot be transferred directly to this individual mixed-objective variant. No monotonic system improvement follows from a local surrogate increase. PPO clipping follows the standard policy-surrogate principle [3].

Optional exact categorical KL stopping, rollback and joint budgets exist but are disabled in investigated baselines. A crossing local step is retained without rollback; enabled rollback restores actor and Adam state. Local KL averages controllable rows; joint KL sums actor contributions averaged uniformly over the same full rollout rows, without control masks or discount weights. These KLs concern command distributions, not physical-release distributions.

Evidence: Equations (19)-(20): R [A3-A5]. Theoretical comparison T [1-3]; optional controls are not described as active defaults.

10. Integration of HAPPO with Reservoir Dynamics

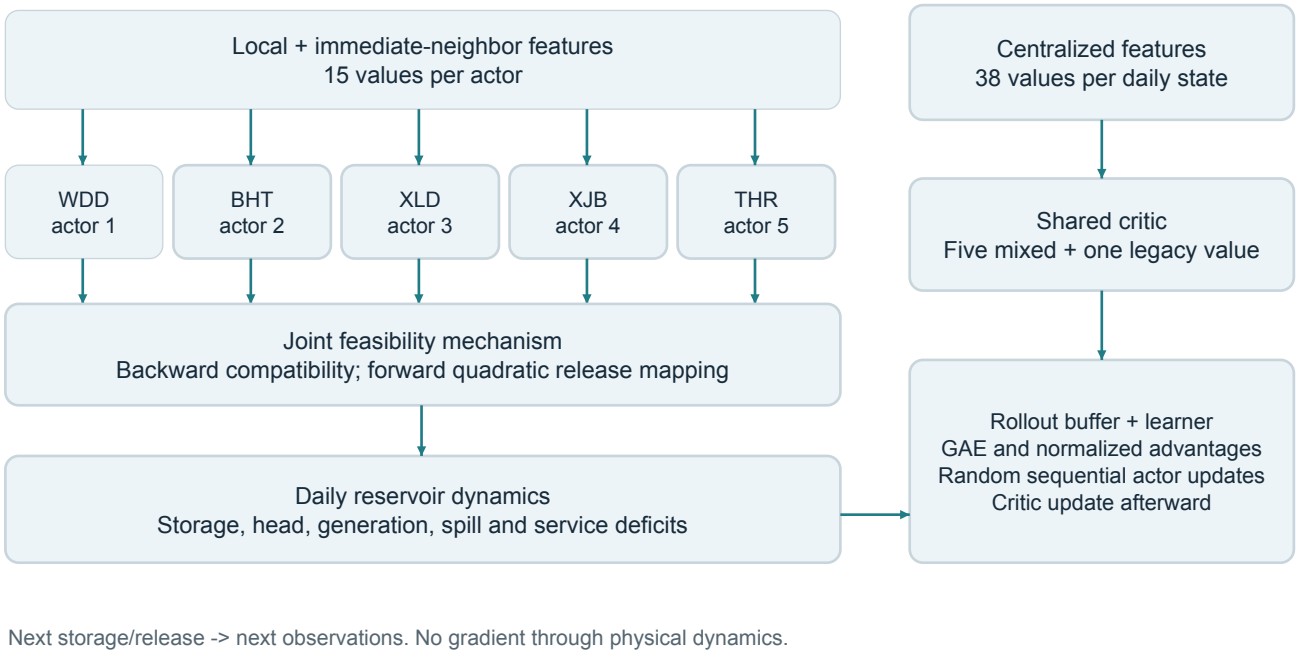


Figure 2. Information, physical execution and learning interaction. Five commands are selected before upstream releases are resolved. The centralized feasibility mechanism mediates their joint physical effect; the critic supplies return estimates during training.

10.1 Actual order of one transition

At day t, actor observations and centralized features are constructed from current storage, today's forcing, previous releases, date and applicable limits. Actors sample commands; the critic predicts six values before execution. Current storage is checked, backward intervals are computed, and total releases are resolved upstream to downstream. Balance then gives next storage; geometry and head give turbine flow, power, spill and energy.

Service deficits and all reward definitions are calculated before the environment commits next storage and actual previous release. The date index advances, termination is checked, and next observations are constructed. A terminal observation is zero. The runner stores transitions, rollout probabilities, masks and values. At a year join no reset occurs.

10.2 Distinct forms of order and coordination

Hydraulic execution follows the fixed river order. Actors select commands using their pre-execution observations. Learning uses a newly randomized actor-update order after complete trajectories are collected. These orders must not be conflated. F changes the weight of stored commands, not the water flow or visited state.

Coordination comes from physical coupling, neighbor information, global generation rewards, centralized value estimation and sequential policy correction. Daily feasibility is also imposed directly by the joint model. No explicit message exchange, counterfactual credit assignment or differentiable physical control layer is implemented.

Evidence: V [M1-M3, A2-A4, P5]. Figure 2 is a conceptual reconstruction, not a diagram of software classes.

11. End-to-End Training and Evaluation Methodology

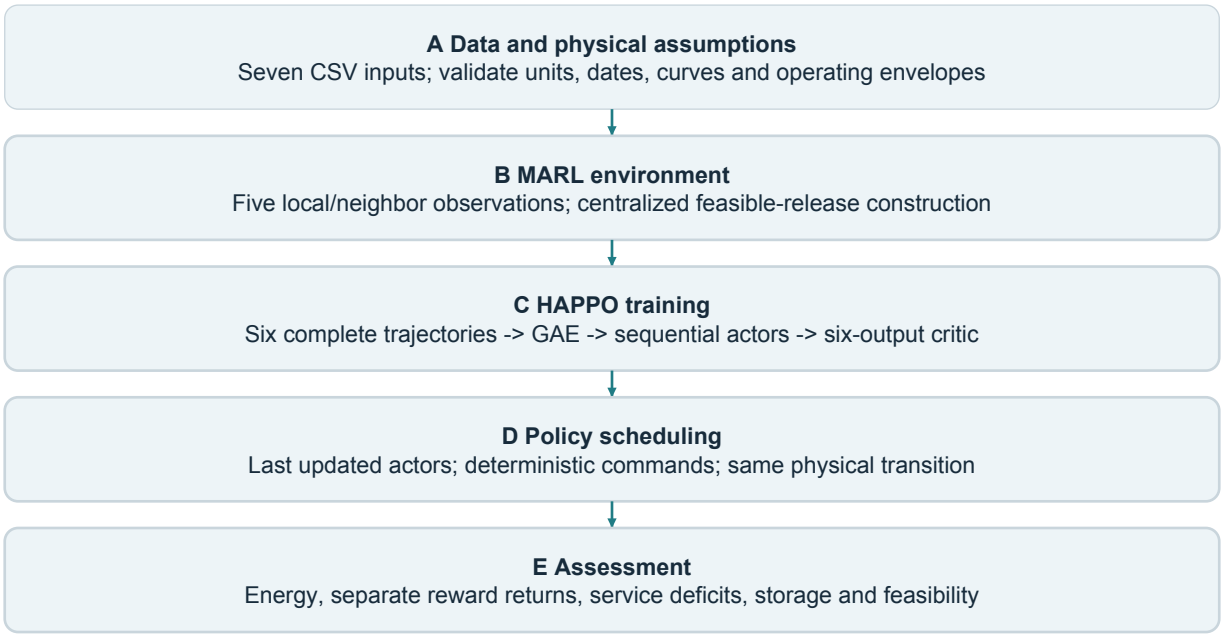


Figure 3. End-to-end scientific workflow. Training contains a collection/update feedback loop. Periodic training-window evaluation and final held-out assessment observe policies; neither supplies a best-score model-selection loop in the implemented workflow.

11.1 Data and episode preparation

Source tables establish geometry, capacities, head assumptions, daily forcing and permitted end-state envelopes. Split checks define admissible consecutive training windows. Each reset loads the start year's initial storage and clears previous release. Input validation does not establish future feasible operation for every possible action history.

11.2 Training feedback and termination

A seeded schedule approximates equal source-year-day exposure with distinct starts per six-lane batch. Actors remain fixed during complete collection. Returns and advantages are computed once; actor updates and then critic updates produce the next policy. The process ends at the fixed episode budget, not at a convergence criterion or an evaluation-score threshold.

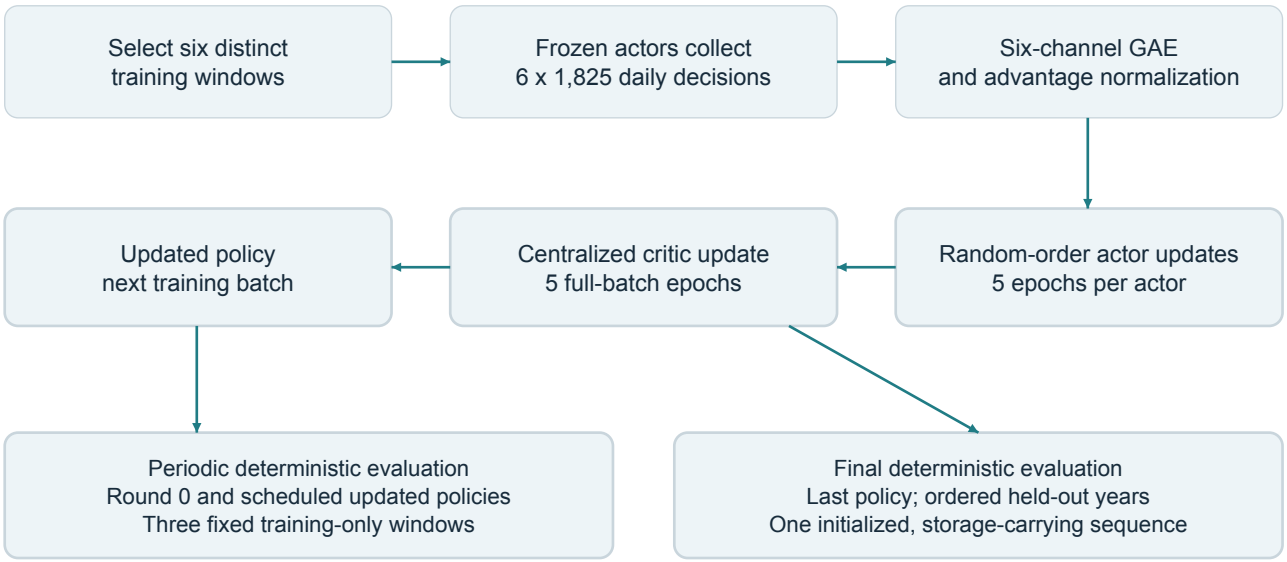
11.3 Deployment and assessment

The last policy generates deterministic daily schedules under the same physics and scaling. Assessment computes energy, separate reward returns, service frequency/severity, storage and numerical feasibility. Optional inference checkpoints enable later frozen-policy replay with strict matching of required settings and fingerprints.

Training/evaluation failure handling is part of the methodology: a failed collection batch aborts without optimizing its partial data. A physical failure in one periodic window invalidates the round aggregate; surviving windows are not silently reweighted. Final physical failure is logged and re-raised, aborting final evaluation without a valid full-horizon score. Periodic physical failures permit training to continue.

Evidence: V [D2-D3, W1-W4]. New training/evaluation was not run for this reconstruction.

11. Training and Evaluation: Time Scales and Policy Use



Evaluation trajectories do not enter training buffers; no best-score policy selection.

Figure 4. Training and evaluation process. Each update follows six complete frozen-policy trajectories. The diagram's evaluation branches are observational and do not feed experience back into the optimizer.

Time scale	Operational meaning	Default count
Physical time step	One daily decision in all five reservoirs	86,400 s; 24 h generation.
Scheduling episode	One initialized storage-carrying trajectory	1,825 daily transitions.
Rollout batch	Six complete parallel episodes	10,950 environment rows.
Optimization round	Sequential actors then shared critic	Five epochs each; one minibatch.
Training per seed	Fixed completed episode budget	300 episodes; 50 rounds; 547,500 days.
Periodic assessment	Untrained and updated frozen policies	Round 0 plus 50 scheduled rounds.
Final assessment	Last policy; ordered held-out years	One sequence per seed.

Table 12. Physical, reinforcement-learning and optimization time scales. A simulated day is one joint five-reservoir transition, not five independent environment samples.

Periodic dry/median/wet windows start in 2007, 1991 and 1973. They are ranked by mean total external inflow across eligible windows, not by held-out year labels. All contain 1,825 days and end on 30 December of their fifth year. Round-k training curves describe stochastic collection under the pre-update policy; round-k evaluation describes the updated deterministic policy.

Final evaluation uses [1998,1999,1970,1997,2006], initializes once from 1998 and carries storage across artificial joins. It is neither a continuous historical period nor five independently initialized annual tests. Saved policies are selected by budget completion, not best validation performance.

Evidence: V [W1-W4]. Scenario classification, rollout accounting and final ordering are independently traceable.

12. Experimental Configuration and Performance Assessment

12.1 Version-aware registry of inspected experiments

ID	K	γ	G/F scales	XLD base / effective MW	Initialization / version qualification
A	200	1	.5/.7	5345.7 / 2672.85	Older source; periodic global-max start.
B	200	1	.4/.4	5345.7 / 2138.28	Older source; changed bounds fingerprint.
C	200	1	.4/.4	5011.6 / 2004.64	Different XLD base; older source.
D	100	.99	1/1	5345.7 / 5345.70	Current year-specific initial vectors.
E	100	.99	.4/.4	5345.7 / 2138.28	Current audited source/data match.

Table 13. Archived effective configurations. All five record seeds 1,2,3; 300 episodes/seed; six workers; 50 updates; coverage-balanced sampling; λ=.95; five actor/critic epochs; and disabled KL guards.

Archive identifiers under runs/:
A: five_year_bins200_seeds123_run1.jsonl
B: five_year_bins200_g40_e40_xld_seeds123_run1.jsonl
C: five_year_bins200_g40_e40_xld_seeds123_run3.jsonl
D: happo_seeds123_20261103_154615.jsonl
E: happo_seeds13_20261103_171312.jsonl

A-C use global-maximum periodic initial storages [5.863,19.006,11.570,4.977,39.300] × 10^9 m^3. D-E use year-specific vectors. Their source fingerprints also differ; C changes an underlying requirement and B has a different dated-bound fingerprint. The archives are not a clean single-factor experiment sequence.

E matches all 54 recorded source fingerprints and seven input fingerprints of the audited working tree. D matches all seven inputs and differs only in environment YAML among its 54 recorded source files. This verifies source/input correspondence, not successful re-execution. The filename seeds13 does not imply only two seeds.

12.2 Assessment conventions and unavailable comparisons

Reported means and sample standard deviations use the three seeds. Periodic return/energy summaries first average three scenarios within each seed; violation rates pool eligible counts within that seed. These are descriptive statistics, not confidence intervals. Source days, overlapping windows and reservoir-days are correlated.

No same-condition rule-based, mathematical-programming, single-agent or alternative-MARL baseline was identified in the investigated training path and five archives. No optimality gap is available. An untrained network at round zero is not a competitive engineering benchmark.

Evidence: V [X1-X2]. Registry settings come from line 1 metadata, not archive names or current defaults.

12. Performance Assessment: Archived Scheduling Outcomes

ID	Energy TWh	Legacy return	G violations %	F violations %	Depletion 10^9 m³
A	1172.735 ± 9.098	640.327 ± 6.694	20.581 ± .601	.8219 ± .0290	39.453 ± 1.200
B	1170.347 ± 5.945	693.788 ± 4.429	9.545 ± .538	.0767 ± .1144	40.074 ± .275
C	1179.726 ± 8.189	704.188 ± 7.077	8.486 ± .799	.0110 ± .0110	38.974 ± 1.227
D	1180.863 ± 6.713	4.558 ± 2.880	49.607 ± .225	9.374 ± .252	33.481 ± .632
E	1185.241 ± 7.757	35.122 ± 1.803	9.724 ± .196	0 ± 0	33.167 ± .720

Table 14. Final held-out sequence: mean ± sample SD across seeds 1-3. Energy, violation frequencies and depletion are undiscounted; returns use each run's γ and effective thresholds. All 15 records report feasible=true and zero daily storage violations.

Returns across A-C and D-E have different discount scales. G/F percentages also use different service definitions. Lower violations at 40% thresholds do not demonstrate compliance with full reference requirements. No causal algorithmic improvement or statistical superiority follows from this table.

E round	Legacy return	Energy TWh	G violations %	F violations %
0	7.906 ± 9.376	1137.378 ± 23.319	24.955 ± 1.321	14.844 ± 2.134
1	15.604 ± 8.962	1167.298 ± 9.959	22.128 ± 1.146	8.745 ± 3.528
10	32.781 ± 1.887	1233.481 ± 5.035	10.194 ± 1.430	.2033 ± .0811
25	34.787 ± 1.490	1264.028 ± 13.504	9.650 ± .433	.00487 ± .00844
50	35.178 ± 1.703	1260.123 ± 7.670	9.679 ± .091	.00244 ± .00422

Table 15. Current E periodic evaluation, three fixed training windows per seed. D at round 50 has energy 1256.332 ± 6.296 TWh, legacy 7.076 ± 2.784, G violations 48.056 ± .112% and F violations 11.183 ± .307%, under its full reference thresholds.

From E round 25 to 50, system return increases from 108.119 ± 4.638 to 109.177 ± 5.035 and every mean actor mixed return rises, while energy decreases and G violation frequency rises slightly. The record illustrates disagreement among criteria; it does not show uniform monotonic improvement.

E's held-out net depletion is about 90.7% of initial storage above global minimum. This is a storage-accounting ratio, not water-particle tracing or attribution of generation to depleted water. Terminal treatment must be considered before interpreting energy as sustainable operating performance.

Evidence: Archived observations V [X1-X2]. Exact records and statistical procedures are listed in Appendix D.

12. Performance Assessment: Reward and Learning Diagnostics

Scope	Generation reward	G penalty	F penalty	Legacy return
D round 0	32.1714 ± 1.4980	30.8676 ± 6.3798	15.5055 ± 5.2064	-14.2016 ± 12.5613
D round 50	35.1742 ± 1.4097	26.7743 ± 1.3104	1.3234 ± .0814	7.0764 ± 2.7845
D held-out	35.2669 ± 1.6587	29.3259 ± 1.0812	1.3826 ± .1834	4.5584 ± 2.8798
E round 0	32.1714 ± 1.4980	14.6066 ± 4.7341	9.6588 ± 3.7769	7.9060 ± 9.3755
E round 50	35.9548 ± 1.6434	.776463 ± .061943	.000039 ± .000067	35.1783 ± 1.7032
E held-out	36.1277 ± 1.7195	1.005382 ± .084802	0 ± 0	35.1223 ± 1.8031

Table 16. Discounted legacy decomposition, γ=.99; mean ± sample SD. Generation minus penalties reconciles per archived record to approximately 10^{-13}. Scalar magnitudes are not gradient-contribution measurements.

Substantial guarantee penalties reduce D's legacy return. E's final return is not penalty-dominated. Identical round-zero energy in D and E alongside different rewards demonstrates the effect of changing service criteria; post-training trajectories also differ.

Existing diagnostic	Observed record	Interpretation limit
D-E actor learning	750 actor updates each; positive parameter displacement; no missing gradients	Parameter changes do not establish useful cooperation.
D-E own-channel fit	750/750 post-update MC RMSE below pre-update in each run	Changes include network and ValueNorm effects on the same batch.
D seed1 round50	Legacy GAE EV .9822; realized-MC EV .8740	Good λ-return fit is not equivalent to accurate expected values.
D seed1 round45 BHT	KL .02234; clip fraction63.315%; surrogate delta -7.98×10^-5	An isolated fixed-batch decrease, with no active KL limit.

Table 17. Selected recorded diagnostics. Realized Monte Carlo return is not the true expectation; raw GAE/MC advantage sign disagreement is not a gradient-error rate. Rowwise sequential-factor concentration was not retained in the examined summaries.

Forced training decisions number 59 in D and 62 in E out of 8,212,500 opportunities per run; all occur at XJB. Saved D final-policy traces have no forced records. The archived D same-day diagnosis often finds a service-meeting alternative, but its storage consequence prevents interpreting independent alternatives as a better multiyear trajectory. Details and arithmetic checks appear in Appendix D.

Evidence: V [X2-X4, A7]. No learning diagnostic was generated by a new experiment in this audit.

13. Prioritized Methodological Diagnosis of Current Performance

Table 18a. Priorities P1-P3 indicate research attention. Confidence concerns evidential support, not the size or causality of performance effects. All verification and improvement directions are future work.

Potential issue	Repository evidence	Methodological explanation	Possible impact	Evidence classification	Confidence	Recommended verification	Potential improvement
P1: Learning versus reported objective	[M3, X2]: five mixed rewards; separate system/legacy returns.	Own utilization and repeated global terms do not equal the displayed scalar or total energy.	A better learning return can accompany worse energy or service frequency.	Evidence-Supported Concern	High	Declare the engineering criterion; rescore the same trajectories under fixed definitions.	Choose an explicit common or individual objective; retain separate physical metrics.
P1: Long-horizon discount alignment	[A3,A6]: γ=.99; explicit γ^t actor weights.	97.448% of pure time-weight mass falls in the first 365 days; this is not gradient mass.	Weak relative weighting of distant operating periods if all days should matter equally.	Evidence-Supported Concern	High	Compare stated objective with discount sensitivity using matched scenarios and budgets.	Select discount/average-reward criteria consistent with the scheduling purpose.
P1: No terminal water valuation	[P6,M3,X2]: no salvage/equality; final net drawdown.	The finite return gives no explicit value to water remaining after the episode.	May favor end depletion over sustainable carryover operation.	Evidence-Supported Concern	High	Evaluate common terminal conditions and energy with net storage change.	Justify terminal bounds, continuation value or longer rolling operation.
P1: Uncalibrated hydraulic fidelity	[P3-P4]: assumed head slope/anchor; constant efficiency.	Conservation and capacity checks do not identify true head, losses or routing.	Optimized modeled energy may differ from real plant operation.	Evidence-Supported Concern	High	Obtain measured levels, tailwater, flow and power; validate routing/time aggregation.	Replace assumptions with calibrated, independently assessed physical relationships.

Evidence: Matrix classifications distinguish confirmed issues, supported concerns and unverified hypotheses. Modeling assumptions are not automatically defects. Source IDs resolve in Appendix B-C.

13. Methodological Diagnosis: Evidence Matrix 2

Table 18b. Priorities P1-P3 indicate research attention. Confidence concerns evidential support, not the size or causality of performance effects. All verification and improvement directions are future work.

Potential issue	Repository evidence	Methodological explanation	Possible impact	Evidence classification	Confidence	Recommended verification	Potential improvement
P2: Observation sufficiency	[M2]: neighbor features, current means, no forecast or recurrent state.	Actors must infer future value from storage, season and learned statistics.	Difficult anticipation of refill, drawdown and distant cascade effects.	Evidence-Supported Concern	Moderate	Test forecast/history information on held-out windows; assess day-mean availability.	Add operationally available predictive/context features after leakage checks.
P2: State-dependent discrete releases	[P5,M1]: quadratic K-bin interval positions.	Same index has changing flow meaning; resolution varies with state and command.	Exploration and upper-interval precision can be difficult.	Evidence-Supported Concern	Moderate	Quantify physical grid gaps and service sensitivity on visited trajectories.	Compare mappings/resolutions under fixed data, objective and budget.
P2: Daily versus future feasibility	[P5-P6]: compatibility uses only current forcing/bounds.	A feasible present decision need not preserve favorable later operating options.	Possible future scarcity, low head or later constraint pressure.	Evidence-Supported Concern	High	Assess multiday reachability and downstream storage consequences.	Explore justified look-ahead feasibility or terminal/continuation formulations.
P2: Shared legacy critic influence	[A1,A5]: six outputs share trunk and value loss.	Legacy target adds representation gradients though actors use own advantages.	May help or compete with mixed-channel fitting; direction unknown.	Evidence-Supported Concern	Moderate	Ablate legacy head with matched seeds; compare each channel and scheduling metrics.	Retain or separate auxiliary targets based on controlled evidence.

Evidence: Matrix classifications distinguish confirmed issues, supported concerns and unverified hypotheses. Modeling assumptions are not automatically defects. Source IDs resolve in Appendix B-C.

13. Methodological Diagnosis: Evidence Matrix 3

Table 18c. Priorities P1-P3 indicate research attention. Confidence concerns evidential support, not the size or causality of performance effects. All verification and improvement directions are future work.

Potential issue	Repository evidence	Methodological explanation	Possible impact	Evidence classification	Confidence	Recommended verification	Potential improvement
P3: Sequential-factor concentration	[A4,A7]: F unclipped; examined logs lack row distribution.	Products of changed likelihoods could concentrate weighted samples.	Potential high-variance updates; no observed explosion established.	Hypothesis Requiring Verification	Low	Inspect F quantiles, effective weight concentration and gradient contributions.	Consider monitored update-size controls if concentration is demonstrated.
P3: Harmful policy concentration	[A7]: entropy decreases in saved diagnostics.	Concentration may be useful specialization or premature exploitation.	Possible reduced exploration; low entropy alone is insufficient evidence.	Hypothesis Requiring Verification	Low	Relate entropy to action coverage, repeated-seed outcomes and useful alternatives.	Adjust exploration only after identifying a demonstrated deficiency.
P3: Stale methodological documentation	[D4,M3]: .5/.7 comment versus .4/.4 values; legacy-only prose.	Some text no longer describes the executable criteria.	Misconfigured experiments and mistaken interpretation or reproduction.	Confirmed Implementation Issue	High	Compare effective metadata against all reported settings and reward labels.	Synchronize documentation and archive complete effective criteria.
P3: Experimental attribution and generalization	[X1-X4]: three seeds; overlapping windows; confounded historical runs.	Limited independent scenarios and absent matched baselines restrict inference.	Uncertain robustness, causal improvements and optimality gap.	Evidence-Supported Concern	High	Use matched baselines, seed replication and independently justified hydrological tests.	Predefine metrics, comparisons, uncertainty summaries and generalization protocol.

Evidence: Matrix classifications distinguish confirmed issues, supported concerns and unverified hypotheses. Modeling assumptions are not automatically defects. Source IDs resolve in Appendix B-C.

14. Limitations, Uncertainties, and Future Research Directions

14.1 Limits of what is established

The investigation reconstructs a working scientific formulation from static code, supplied inputs and archived records. It establishes neither calibrated engineering fidelity nor a fresh successful run. Original hydrology and envelope provenance, measured loss/tailwater relationships and externally specified terminal-water preferences remain unresolved. Exact current-day mean inflow availability is an operational assumption.

Common-return HAPPO theory, internally feasible schedules and improving fixed-batch surrogates are different forms of evidence. None establishes that this mixed-objective implementation has found a globally optimal reservoir schedule. The practical centralized features are not proven Markov-sufficient.

14.2 Evidence-based sequence for future investigation

Future stage	Question to resolve	Proposed direction (not implemented)
Objective specification	What constitutes satisfactory operation?	Fix horizon, terminal value, service levels and primary system criterion before tuning.
Physical validation	Does the transition represent the plants?	Authenticate inputs; calibrate/validate geometry, head, efficiency, routing and daily aggregation.
Diagnostic instrumentation	Which learning signal is problematic?	Measure per-term gradients, factor concentration, physical action grids and channel-specific value errors.
Controlled method comparisons	Which choice causes a change?	Matched-seed, same-data ablations of discount, terminal treatment, observations and action mapping.
Operational assessment	Will policies generalize and sustain storage?	Independent hydrological scenarios, rolling operation, matched rule/optimization baselines and uncertainty analysis.

Table 19. Proposed research sequence. Recommendations are hypotheses and verification directions, not claims that the listed changes will improve performance.

14.3 Interpretation boundaries

Threshold relaxation changes the problem; lower violations do not prove algorithmic improvement. A training-window plateau is not convergence. Three-seed SD is not a confidence interval. Same-day service alternatives can consume storage and cannot be concatenated into a better feasible trajectory. Current-state head/water classifications may reflect earlier policy choices rather than unavoidable hydrology.

The report therefore preserves both counterevidence and limits of observation: forced actions are rare in inspected records; current final scalar returns are not penalty-dominated; rowwise factor explosion or concentration could not be assessed from the retained summaries; and no grid-gap category occurred in the archived reference diagnosis. These facts narrow the diagnosis without proving universal adequacy.

Evidence: V/C/H/F as specified; recommendations depend on evidence [P1-P6, M1-M5, A1-A7, X1-X4].

15. Reproducibility Framework and Audit Boundaries

15.1 What must be identified for a scientific rerun

A reproducible study must identify the seven exact input files; split and source-year order; episode horizon; initial-condition rules; daily-envelope timing; head model and assumptions; service base values, scales and weights; observation layout; K and release mapping; all optimization settings; random seeds; and software versions. Effective CLI overrides must be archived with YAML defaults.

The audited revision is 6473ff6263a48e0fa62ba3c10b946fdd036777ad. The root is /Users/hujingxin/Documents/Code_python/HARL-main. Evidence references below are relative to that root and use one-based line numbers. Source identity is version-specific; outputs alone do not identify an experiment.

Artifact / control	Verified role	Reproducibility limit
Run configuration metadata	Effective settings, input/source fingerprints, seed and split	Recorded fingerprint scope is not every external dependency.
Run source hashes	54 files in investigated configuration metadata	Exclude setup.py; checkpoint source scope includes it (55 files).
Inference checkpoint	Five actors, critic, ValueNorm, settings and fingerprints	No complete optimizer/sampler/RNG state for identical training resume.
Frozen-policy replay	Strict source/settings/data checks; saved horizon and physics	No replay was executed for this report; payload behavior unverified here.
Compact step CSV	Fifteen scalar daily summaries, limited precision	Cannot reconstruct a complete five-reservoir schedule.
Full replay / optional trace	Reservoir-level states, commands and learning/physical quantities	Optional traces are not assumed present for every experiment.

Table 20. Recorded reproducibility support. Training failure aborts partial batches; final physical failure does not produce a valid complete-horizon result. These rules must be preserved when comparing successful and failed runs.

15.2 Audit protections and verification scope

Repository inspection remained read-only. No project module, checkpoint, training, evaluation, simulation, data-generation program or test suite was executed. Existing output tables were inspected with general-purpose parsing and arithmetic only. Test presence is not evidence that tests passed.

Before PDF authoring, 2,658 files totaling 5,124,140,326 bytes were fingerprinted, including Git metadata, alongside paths, file modes and modification times. Final delivery verification compares that baseline with the post-generation tree while excluding only the requested PDF. All authoring/rendering intermediates are confined to a task-created temporary directory outside the repository.

Evidence: V [W4, X1, X4]. Final file/layout checks concern document quality and repository protection, not scientific model execution.

16. Conclusions

The repository formulates daily control of five hydraulically coupled reservoirs using prescribed historical incremental inflows, table-based geometry, assumed head relationships and a joint current-day release-feasibility mechanism. Categorical commands select quadratic positions within computed feasible intervals. Water balance, total outlet capacity and modeled generation limits constrain execution. Output and ecological requirements remain reward-based preferences.

Five independent actors learn different local/global mixed returns using a centralized six-output critic, GAE and randomized HAPPO-style sequential updates. Rollouts span full 1,825-day episodes; policy optimization retains explicit episode-start discount weights. Deterministic last-policy evaluation uses both fixed training scenarios and an ordered held-out sequence with storage carryover.

The archived experiments support descriptive performance assessment and internal arithmetic checks. They do not provide a controlled algorithmic comparison, an optimality certificate, calibrated engineering validation or a causal explanation for every unsatisfactory result. Distinct objectives, discounting, terminal-water treatment, information limits and physical simplifications merit attention before attributing performance to a particular optimizer setting.

The resulting methodology is independently interpretable, but its future scientific claims should remain conditional on explicit operating criteria, authenticated/calibrated inputs and controlled evaluation. Proposed improvements in this report are research directions rather than implemented or demonstrated gains.

References

[1] Kuba, J. G., Chen, R., Wen, M., Wen, Y., Sun, F., Wang, J., and Yang, Y. (2022 revision). *Trust Region Policy Optimisation in Multi-Agent Reinforcement Learning.* arXiv:2109.11251. https://arxiv.org/abs/2109.11251. Used for sequential heterogeneous-agent theory, not repository results.

[2] Zhong, Y., Kuba, J. G., Feng, X., Hu, S., Ji, J., and Yang, Y. (2024). *Heterogeneous-Agent Reinforcement Learning.* Journal of Machine Learning Research, 25(32), 1-67. https://www.jmlr.org/papers/v25/23-0488.html.

[3] Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O. (2017). *Proximal Policy Optimization Algorithms.* arXiv:1707.06347. https://arxiv.org/abs/1707.06347.

[4] Xie, Z., et al. (2026). *Shared versus reservoir-specific policy architectures in multi-agent reinforcement learning for physically heterogeneous cascade reservoir operation.* Journal of Hydrology. DOI: 10.1016/j.jhydrol.2026.136160. Publisher record. The title/DOI identity was corroborated; the repository's exact Table A.1/page-14 attribution and imported values were not independently checked against the table. No paper performance result is presented as a repository result.

17. Appendices: Methodological Evidence and Traceability

A. Symbol and equation register

Symbols	Meaning	Units / scope
i; t; T	Reservoir index; executed action-day index; episode length	i=1,...,5; t=0,...,T-1; default T=1825.
V; a,b	Beginning/resulting storage; resulting-state permitted limits	m^3; date-t limits apply after action t.
e; I; Q; Q cap	External inflow; total inflow; total release; outlet capacity	m^3/s.
L^d,U^d; L,U	Downstream-compatible and local feasible release bounds	m^3/s; distinct from local reward ℓ.
q nom; q tur; q spill	Nominal turbine limit; turbine flow; spill	m^3/s.
Z(V); Z ref; H; H ref	Interpolated elevation; assumed reference; net/reference head	m.
ρ; g; η; c	Density; gravity; efficiency; power/flow coefficient	kg/m^3; m/s^2; dimensionless; $MW/(m^3/s)$.
P; P cap; E	Daily mean power; installed capacity; daily energy	MW; MW; MWh.
G; F; dP,dF	Effective service thresholds; relative deficits	MW; m^3/s; dimensionless.
ℓ; g_t; r; R	Local reward; global generation reward; daily reward; return	Dimensionless; distinguished from gravity g.
k; K; u; θ	Categorical command; bin count; normalized command; policy weights	Dimensionless.
o; x; ψ; v	Actor observation; centralized features; critic weights; value	Scaled features; values in return units after denormalization.
δ; Â; Ã; Y	TD residual; GAE; normalized advantage; λ-return target	Return units except normalized Ã.
γ; λ; m; C	Discount; GAE trace parameter; continuation; controllability	Dimensionless; m and C are different masks.
ρ_i,b; F_b; ε; β_H	Policy ratio; sequential factor; clip width; entropy weight	Dimensionless; distinct from water density ρ.
B; b; ν; Nν	Rollout-row count; row index; ValueNorm moments; normalization	B=10950; b combines lane and episode time.

Table A1. Context and subscripts distinguish reused conventional symbols. Main equations define quantities where first introduced. All numbered equations are R; equation (6) is also corroborated by documented scenario formulas. Theory references [1-3] are not substituted for implementation evidence.

$$\begin{aligned} &\mathrm{Rate}_G = n_{\text{shortfall } G}/n_{\text{eligible } G}; \\ &S_P = \sum_{i,t}\max(G_i - P_{i,t}, 0) \times 24; \\ &S_F = \sum_{i,t}\max(F_i - Q_{i,t}, 0) \times 86{,}400. \end{aligned} \tag{21}$$

Equation (21) gives undiscounted frequency and integrated service deficits (MWh and m^3), after the implementation's shortfall tolerance. Ecology frequency is analogous. Inactive requirements contribute no eligible days; no active requirements yields an undefined, not zero, rate.

Evidence: Equation groups: (1) learning criterion; (2)-(5) balance/geometry; (6)-(8) generation; (9)-(11) feasibility/action; (12)-(15) rewards; (16)-(20) learning; (21) metrics [M5].

17. Appendix B: Physical, Data and Decision Evidence

Paths are relative to the audited repository root (Chapter 15), with one-based line numbers. Prefix **E/** = harl/envs/cascade_reservoir/; **R/** = harl/runners/; **C/** = harl/configs/. Main-chapter identifiers point to this ledger.

ID	Claim / methodological role	Source locator	Kind
P1	Connectivity, mass balance, global residual	E/reservoir_physics.py:130-158	V/R
P2	Reservoir capacities and global domains	data/cascade_reservoir_new/reservoir_parameters.csv:1-6; C/envs_cfgs/cascade_reservoir.yaml:3-13	V
P3	Assumed head data and validation	data/cascade_reservoir_new/reservoir_head_sensitivity.md:5-25; E/data_loader.py:206-250	V/R
P4	Mean head, turbine allocation, power and energy	E/reservoir_physics.py:53-84	V/R
P5	Backward/forward release compatibility and forced branch	E/reservoir_physics.py:100-145	V/R
P6	Current/action-date boundaries, reset and terminal	E/cascade_reservoir_env.py:71-92,154-198; E/reservoir_physics.py:42-51	V
D1	Source data validation, training scaler, geometry	E/data_loader.py:84-184,252-274	V
D2	Eligible windows and ordered daily views	E/episodes.py:15-89	V
D3	Sampling quotas and no duplicate starts per batch	E/training_sampler.py:107-204; E/data_loader.py:348-424	V
D4	Service base/scaled thresholds and citation	E/data_loader.py:276-346; data/cascade_reservoir_new/reservoir_service_requirements.csv:1-6; companion .md:3-10,47-53	V; attribution partial
M1	Agent assignment and command space	E/cascade_reservoir_env.py:28-69,145-152	V
M2	Exact 15/38 features and neighbor placeholders	E/cascade_reservoir_env.py:94-143	V
M3	Deficits and all daily reward definitions	E/reward.py:97-153	V/R
M4	Forced masking and six-channel insertion	R/cascade_reservoir_runner.py:221-236	V
M5	Discounted totals and undiscounted deficit metrics	E/reward.py:156-264; E/periodic_evaluation.py:27-45	V/R

Table B1. Source traceability for physical and environment claims. Source statements are checked against executable logic and current data rather than presumed correct from comments. Full numerical curve/envelope counts came from read-only inspection of the supplied tables.

Direct input filenames under data/cascade_reservoir_new/: daily_incremental_inflows_1965_2024.csv; initial_storage_1965_2024.csv; reservoir_stage_storage.csv; reservoir_parameters.csv; reservoir_head_sensitivity.csv; reservoir_service_requirements.csv; five_reservoir_happo_input_bounds_1965_2024.csv.

17. Appendix C: Learning and Workflow Evidence

Prefixes E/, R/ and C/ have the meanings defined in Appendix B. D_dispatch_analysis abbreviates runs/happo_seeds123_20261103_154615_dispatch_analysis/. These locators identify evidence; source names do not organize the report's scientific explanation.

ID	Claim / methodological role	Source locator	Kind
A1	Actor/critic architecture and six value heads	harl/models/base/mlp.py:18-68; harl/models/value_function_models/v_net.py:23-50; R/cascade_reservoir_runner.py:104-114	V
A2	Collection and GAE with terminal mask	R/cascade_reservoir_runner.py:200-280; harl/common/buffers/on_policy_critic_buffer_ep.py:85-129	V/R
A3	PPO, entropy, discount weights and forced masks	harl/algorithms/actors/happo.py:181-265,314-334; harl/common/buffers/on_policy_actor_buffer.py:81-100	V/R
A4	Per-channel normalization and sequential F	R/on_policy_ha_runner.py:33-59,70-145	V/R
A5	Six-channel half-MSE and ValueNorm chronology	harl/algorithms/critics/v_critic.py:89-115,139-187; harl/common/valuenorm.py:10-88; harl/utils/models_tools.py:50-52	V/R
A6	Hyperparameters, budgets and optional KL settings	C/algos_cfgs/happo_cascade.yaml:4-60; harl/algorithms/actors/happo.py:77-105,338-416; harl/algorithms/actors/joint_kl.py	V
A7	Learning diagnostic definitions and observations	E/diagnostics.py:102-170; E/learning_signal.py:17-38,70-154; D/E run records in Appendix D	V; impact C/H
W1	Seed orchestration and complete-batch training	examples/train_cascade_parallel.py; R/cascade_reservoir_runner.py:178-383	V
W2	Fixed scenario ranking and evaluation isolation	E/periodic_evaluation.py:62-145	V
W3	Deterministic scheduling and physical assessment	E/evaluation.py:67-167; R/cascade_reservoir_runner.py:385-427	V
W4	Inference checkpoint, matching, replay and fields	E/checkpoint.py:75-160,176-246,289-388; examples/replay_cascade_checkpoint.py	V
X1	Effective run configurations and fingerprint scope	runs/A-E filenames in Table 13, each line 1; examples/train_cascade_parallel.py; E/checkpoint.py:75-81	V
X2	Final/periodic numerical records and returns	D/E records: Appendix D; A-C final lines1358-1360; E/reward.py:185-246	V/R
X3	Saved counterfactual diagnosis and reconciliation	analysis/dispatch_diagnostics/feasibility.py; D_dispatch_analysis/findings.md, report.md, summary.json, manifest.json	V; scoped
X4	Archive arithmetic and fingerprint inspection	12 existing D_dispatch_analysis/*_dispatch.csv files; manifest.json:2-39,3936; read-only arithmetic described in Appendix D	V/R

Table C1. Algorithm, workflow and archive traceability. The existence of diagnostic or test code does not establish execution or passage. Optional functionality is distinguished from defaults and recorded active settings.

17. Appendix D: Archived Results and Numerical Audit

D.1 Exact records and aggregation

For both D and E, round 0 seed lines are 3/456/909; round 1: 12/465/918; round 10: 93/546/999; round 25: 228/681/1134; round 50: 453/906/1359. Final held-out records are 1361-1363 and run_complete is 1364. A-C finals are 1358-1360 and completion is 1361. D seed1 learning diagnostics are line 11 (update1), 407 (update45), and 452 (update50); E update50 seed lines are 452/905/1358.

A mean and sample SD are computed across three seeds. Periodic scenario rewards/energy are averaged within seed first; service frequency pools violation and eligible counts over its three windows. Averaging these same-sized windows and pooling counts agree only when eligibility is identical. No independence of reservoir-days is assumed.

D saved diagnosis	Violations / eligible	Frequency	Same-day discrete meeting share
Fixed windows: guarantee	39,466 / 82,125	48.056%	68.575% of violations
Held-out: guarantee	13,580 / 27,375	49.607%	68.130% of violations
Fixed windows: ecology	9,184 / 82,125	11.183%	99.456% of violations
Held-out: ecology	2,566 / 27,375	9.374%	100% of violations

Table D1. Existing reference-run diagnosis. Among guarantee violations, fixed-window head/water shares are16.282%/15.142%, and held-out shares 17.172%/14.698%. No grid-gap, capacity or dispatch-only category appears. These outcomes concern this model and visited states only.

Alternatives fix the saved beginning state and other actors' commands, not their actual releases: the joint map may change those releases. Each alternative is a separate same-day counterfactual. They do not advance a counterfactual trajectory and cannot establish superior multiyear policies.

D.2 Read-only saved-trajectory consistency checks

Audit of 12 saved D CSVs	Observed result
Size / ordering	12 × 9,125 rows = 109,500 reservoir-days; 21,900 environment-days; required fields and ordered five-node/day groups checked.
Maximum balance residual	Single reservoir 3.5763×10^-6 m³; cascade 4.7684×10^-6 m³.
Propagation / flow partition	Maximum I_i-e_i-Q_(i-1): 3.6380×10^-12 m³/s; Q-turbine-spill: 0.
Forced decisions / narrowest interval	0 forced rows; minimum width approximately 97.29 m³/s.
Archive reconciliation	Run-log and all 12 CSV hashes match manifest;12 recorded scenario matches have maximum difference0.

Table D2. Arithmetic checks recompute relationships from saved numbers; they are not a fresh physics/replay execution. The manifest's recorded replay match is archive evidence, not an evaluation performed for this report.

The referenced summary.json phase/service section begins at line 405268 and reward accounting at4007. The manifest reconciliation begins at38 and total-row record is3936. Saved D analysis does not automatically diagnose E. Exact paper-table attribution, physical calibration, causal improvements and cross-platform/runtime reproduction remain unresolved.

17. Appendix E: Evidence Status and Input Identity

Input under data/cascade_reservoir_new/	SHA-256 at final authoring
daily_incremental_inflows_1965_2024.csv	265c16c233e74d0be2d309b5b01bfc3cafb7a63cefe98c8df6266c73f63373ef
initial_storage_1965_2024.csv	b9e292c77bb72ae5c076354fc3ab044fd66211181da195dcc6eb6747e4d7d9d4
reservoir_stage_storage.csv	86f1ba09cee090002bc349a8d97377c76a94c1a782f5c589d2e2d0bea0411988
reservoir_parameters.csv	095a9ff1a0486ab716b069895370120e40304a72d06ff0c81d50e301104a528c
reservoir_head_sensitivity.csv	c1cd6c2836aa676a1ade850416359337518d51fb56b2e0dc2827980c629a838f
reservoir_service_requirements.csv	6e859bace3f14d7260be788286540cc184a511250f3ef8fec67929f0a2c310d2
five_reservoir_happo_input_bounds_1965_2024.csv	7e70c1da96fb901296c5550d8ca25dbe27413c2047ae1aeeface5b150c26f852

Table E1. Input fingerprints identify the actual supplied files; matching a fingerprint does not authenticate original provenance. Corresponding archived source/input maps are retained in run metadata [X1].

Evidence status	Established scope / explicit boundary
Fully verified	Current model equations, action/reward definitions, feature layout, update chronology, defaults and archived numerical records.
Partially verified	The service paper's bibliographic identity is corroborated; exact Table A.1 values/page14 attribution was not independently inspected.
Conflicting documentation	YAML .5/.7 prose versus .4/.4 values; service prose describes legacy reward; README equality-bound statement absent from current table.
Unresolved provenance / calibration	Original hydrology, initial-storage and envelope generation; curve provenance; measured head/loss/efficiency/routing validation: Not identifiable from the available repository.
Unestablished performance claims	Optimality, baseline superiority, causal effects of proposed changes, Markov sufficiency and broad out-of-sample generalization.

Table E2. Verification status is separate from issue classification. A verified modeling assumption is not automatically a defect; a plausible performance mechanism is not a demonstrated cause.

The pre-generation aggregate repository fingerprint is b8f266c802094921407e20bf9be9794352e4c4855d6e870b9294f804cf5bb19d. It covers a canonical sorted manifest of relative paths, byte hashes, sizes, modes and file modification times, plus directory modes and symlink targets. The final comparison excludes only HAPPO_Five_Reservoirs_Methodology_Report.pdf.

This document contains the methodological reconstruction and evidence limits in full. No project execution, package installation or scientific experiment was performed to fill evidence gaps. Proposed checks and improvements are explicitly future work.